Character-Filtered Exact Cleaners and Galois-Structured Design for Large-Prime BGV Bootstrapping

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Abstract. Large-prime BGVbootstrapping recently became practical by combining bounded support with null-polynomial reductions over $\mathbb{Z}_{p^e}$. This paper studies the next structural layer inside that improved pipeline. We view the nonlinear cleaner and the surrounding homomorphic linear transforms through a common Galois- and character-structured lens. On the nonlinear side, for a fixed bounded support $S_A = \{hA + \ell \bmod p : h, \ell \in [-B, B]\}$, the exact cleaner $P_A(hA + \ell) = \ell$ has a unique degree-$< |S_A|$ representative in the support quotient. Hence, after bounded-support degree reduction, the stable remaining objective is not a lower degree for the same support, but a lower encrypted polynomial-evaluation cost. Our first contribution is to show that this cost can be reduced by selecting auxiliary radices whose support geometry induces a small character space.
For the target parameter set $p = 65537$, $B = 17$, we find and certify the order-four radix $A = 256$, satisfying $A^2 = -1 \bmod p$. This radix makes the support stable under a quarter-turn and forces the canonical cleaner to satisfy $P_A(AX) + AP_A(X) = AX$. Equivalently, the monomial basis of the support quotient decomposes into finite-field character spaces, and all exponents except X and $k \equiv 3 \pmod 4$ vanish. The resulting degree-minimal exact cleaner keeps the same minimal degree 1223 as the generic baseline radix $A = 35$, but its nonzero monomial count drops from 612 to 307. We implement this order-four cleaner and its specialized evaluator in an isolated HElibartifact. Real ciphertext thin bootstrapping preserves correctness and reduces digit extraction from 160.783 s to 80.770 s and full thin bootstrapping from 213.103 s to 131.181 s, speedups of 49.76% and 38.44% over the direct large-prime baseline.
Our second contribution is a conservative generalization framework. We show how higher-order subgroup structure yields an order-adaptive cleaner planner: for $p = 65537$, order four is the stable global choice at $B = 17$, while order eight only becomes support-safe once the local support radius drops to $B \leq 7$, and order sixteen only for tiny local supports. These claims are supported by exact offline interpolation and evaluation-cost sweeps, not by a new ciphertext implementation. Our third contribution is a Galois-structured view of the linear-transform side. Using the square-free cyclotomic index $m = 50731 = 97 \cdot 523$, we explain how normal-basis and character-factorized transform designs turn the dominant EvalMap block into a group-matrix problem; for the $D = 96$ block, a plaintext-exact Rader/FFT factorization reduces the operation count

to 0.703× the dense primitive-Vandermonde path. The current ciphertext integrations, however, remain negative or engineering-only evidence, so the paper states them as design guidance and claim boundaries rather than as a second demonstrated speedup.

Keywords: Fully homomorphic encryption · BGV bootstrapping · Null polynomials · Digit extraction · Exact polynomial synthesis · Galois structure · Homomorphic linear transforms · HElib

1 Introduction

BGVand BFVare standard second-generation fully homomorphic encryption schemes for exact arithmetic. They support SIMD-style packed evaluation over finite rings and are used in private database queries, exact finite-field computation, and other workloads where approximate CKKS-style semantics are not acceptable. Their main remaining cost is bootstrapping: once leveled homomorphic evaluation consumes the modulus chain, the scheme must homomorphically refresh the ciphertext by evaluating a decryption circuit.

In the HElibbootstrapping blueprint, this decryption circuit has two dominant kinds of operations. First, homomorphic linear transformations move between the slot and coefficient representations. Second, a digit-extraction or error-cleaning polynomial removes the encrypted low-digit overflow term. Halevi and Shoup substantially optimized the linear maps through diagonal matrix decompositions, baby-step/giant-step (BSGS), and hoisting [6]. On the nonlinear side, Chen and Han improved lower-digit removal [1]; Geelen, Iliashenko, Kang, and Vercauteren used polynomial-function representatives over $\mathbb{Z}_{p^e}$ [3]; and Ma, Huang, Wang, and Wang showed that in the large-prime regime the cleaner input lives on a small bounded support, enabling much lower-degree representatives through global and local null polynomials over $\mathbb{Z}_{p^e}$ [9].

The large-prime case is the motivating regime here. A common exact plaintext prime is $p = 65537 = 2^{16}+1$, but classical digit extraction has a cost growing with p or $\sqrt{p}$. The bounded-support large-p line dramatically reduces that cost by changing the function domain from the full ring to a small support set. This raises the next natural question: once the degree has already been reduced by support-aware null-polynomial methods, what remains to optimize?

This paper argues that the next stable optimization layer is **structural**, not another attempt to beat the fixed-support degree minimum. For an auxiliary radix A, define the bounded support

$$S_A = \{hA + \ell \bmod p : h, \ell \in [-B, B]\}$$

and the exact cleaner

$$C_A(hA + \ell) = \ell.$$

If the map $(h, \ell) \mapsto hA + \ell$ is injective on $[-B, B]^2$, then there is a unique degree-$< |S_A|$ polynomial P_A that represents C_A on that support. Therefore, after support reduction, one cannot generally claim a lower-degree polynomial for the same fixed support unless the canonical representative already has it. The remaining room is to choose a support geometry whose canonical representative is easier to evaluate homomorphically.

Our first contribution is to show that the auxiliary radix controls this support geometry in a highly algebraic way. The implemented case is $A = 256 \in \mathbb{F}_{65537}^{\times}$, which satisfies $A^2 = -1 \bmod p$. Then multiplication by A acts on the support coordinates as the quarter-turn

$$(h, \ell) \mapsto (\ell, -h).$$

Because the cleaner extracts the low coordinate, this symmetry implies

$$P_A(AX) + AP_A(X) = AX.$$

In the monomial basis, the substitution operator $X \mapsto AX$ is diagonal with eigenvalue A^k on X^k, so the cleaner is forced into a tiny character subspace: only the monomial X and exponents $k \equiv 3 \pmod 4$ survive. Experimentally, the term count drops from 612 for the generic baseline radix $A = 35$ to 307 for $A = 256$, while the degree stays at the same minimum value 1223. In a real HElibartifact, the sparse cleaner plus its specialized order-four evaluator reduces extraction from 160.783 s to 80.770 s and the full thin bootstrap from 213.103 s to 131.181 s.

Our second contribution is a conservative generalization of this viewpoint. Because $(\mathbb{Z}/p\mathbb{Z})^{\times}$ is cyclic, the order-four case is only the first nontrivial subgroup-structured cleaner. For $p = 65537$, the dyadic tower in $(\mathbb{Z}/p\mathbb{Z})^{\times}$ suggests order-8 and order-16 radices as well. However, higher subgroup order is useful only if the support representation remains collision-free. We therefore formulate an order-adaptive planner: the cleaner order is chosen jointly with the support radius. The resulting threshold table is sharp enough to guide the paper's claim scope: for $p = 65537$, order four is the stable global choice at $B = 17$; order eight becomes support-safe only for smaller local supports such as $B \leq 7$; and order sixteen is restricted to tiny local supports. These are exact finite-field and evaluation-cost results, not implemented ciphertext speedups.

Our third contribution concerns the linear-transform side. The same Galois structure that filters cleaner monomials also governs the slot algebra and EvalMap matrices. For the target square-free cyclotomic index $m = 50731 = 97 \cdot 523$, the slot dimensions are determined by $\mathrm{ord}_m(p)$, yielding a dominant $D = 96$ block and a smaller $D = 29$ block. Under a normal-basis / group-matrix viewpoint, these transforms are no longer arbitrary dense linear maps but exact character-factorized group transforms. Offline planner evidence shows that the $D = 96$ block is the real structured target: its HElib BSGS estimate uses about 18 automorphism classes, whereas a factorized mixed-radix plan uses about 7, and a primitive-length Rader reduction yields about 9; exact plaintext verification confirms the $\ell = 97$ block and reduces the operation count to $0.703\times$ the dense primitive-Vandermonde path. The current ciphertext implementations, however, either fail the noise/setup budget or remain purely engineering optimizations. We therefore present the transform side as a mathematically grounded design route and as a co-design principle, not as a second positive ciphertext speedup claim.

This leads to the paper's unifying thesis:

$$\text{bootstrapping cost} = \text{linear-transform structure} + \text{cleaner structure},$$

and the two are coupled. A transform can reshape support geometry and local bounds; those bounds determine which cleaner order is feasible; and the cleaner order determines which character space survives and how cheap the encrypted polynomial evaluation can become. The current order-four implementation is the first stable point in this broader Galois-structured design space.

Claim boundary. The paper's quantitative claims are intentionally asymmetric. The positive ciphertext claim is specific to a fixed HEliblarge-prime BGVthin-bootstrapping pipeline with the implemented order-four cleaner and evaluator. Higher-order cleaners and Galois-structured linear transforms are presented as exact theory, offline planner evidence, exact plaintext verification, and negative/guardrail ciphertext experiments. The paper does not claim a globally optimal bootstrapping pipeline or a demonstrated ciphertext speedup from the new linear-transform algebra.

Contributions. We separate the paper's claims into evidence-backed technical units.

1. **Character-filtered exact-cleaner formulation.** We formulate bounded-support large-prime cleaning in the support quotient and show that, after fixed-support degree reduction, the stable optimization target is sparsity and encrypted evaluation cost rather than a lower degree for the same support.
2. **Implemented order-four cleaner and evaluator.** We prove the order-four cleaner identity $P_A(AX) + AP_A(X) = AX$ for $A^2 = -1 \bmod p$, derive the support restriction $P_A(X) = \frac{1}{2}X + X^3Q(X^4)$, and implement this path in an isolated HElibartifact. The repeat-three ciphertext benchmark gives a 49.76% extraction speedup and a 38.44% full thin-bootstrapping speedup over the direct large-prime baseline.
3. **Order-adaptive cleaner theory.** We give a conservative higher-order generalization based on subgroup action, signed-circulant support models, and support-safety thresholds. For $p = 65537$, the resulting planner shows that order four is the stable global choice at $B = 17$, order eight becomes feasible only once the support radius is sufficiently smaller, and order sixteen is reserved for tiny local supports.
4. **Galois-structured linear-transform design route.** We reinterpret the main EvalMap blocks through the cyclotomic Galois group and a normal-basis / character-factorized matrix model. For the target $m = 50731 = 97 \cdot 523$, we identify the $D = 96$ block as the promising structured target, support that claim with exact plaintext Rader/FFT verification, and explain why the $D = 29$ block is not the first optimization target.

5. **Co-design evidence and claim boundary.** We report the existing negative and engineering-only linear-transform results, use them to delimit the paper's scope, and show how transform support shaping and feasible cleaner order fit into a single planner objective without overclaiming a second positive ciphertext route.

Roadmap. Section 2 positions the work against HElibbootstrapping, null-polynomial digit extraction, and recent BGV/BFV/CKKSalternatives. Section 3 gives the pipeline overview. Section 4 develops the support quotient and fixed-support degree certificate. Section 5 introduces the character-filtered radix structure, with order four as the implemented case and order-adaptive higher-order thresholds as the conservative generalization. Section 6 describes the exact search and planner pipeline. Section 7 records the artifact. Section 8 separates positive ciphertext results, order-adaptive planner evidence, and structured-transform design evidence. Section 9 summarizes limitations and the co-design outlook.

2 Background and Related Work

2.1 HElib Bootstrapping

HElibimplements packed BGVover cyclotomic rings. At a high level, a ciphertext encrypts slots in a plaintext algebra, and bootstrapping evaluates the decryption map homomorphically. The thin bootstrapping path used in this paper performs two families of operations:

– **Linear transformations.** Slot-to-coefficient and coefficient-to-slot transforms are implemented as sums of automorphisms multiplied by diagonal constants. Halevi and Shoup's faster linear transformations use BSGS and hoisting to amortize key-switching and automorphism costs [6].
– **Digit extraction.** A nonlinear polynomial removes lower digits or cleans the encrypted error term so that the refreshed ciphertext decodes correctly.

The current work keeps the mature HEliblinear maps in place. We attempted level-conserved and fused key-switch variants inspired by recent LCR-AKS code explorations, but the correct variants were slower and several aggressive variants failed ciphertext correctness. Those results are reported as negative evidence in Section **??**.

2.2 Digit Extraction and Null Polynomials

Chen and Han introduced homomorphic lower-digit removal polynomials that reduced the depth of digit extraction [1]. Geelen, Iliashenko, Kang, and Vercauteren then studied polynomial functions modulo p^e and used the resulting representative freedom to improve bootstrapping [3]. The key algebraic idea is that a polynomial function over a finite ring can have many polynomial representatives; their differences are null polynomials.

Ma et al. made this perspective practical for large plaintext primes by exploiting bounded support [9]. The input to the relevant digit-removal step is not arbitrary in $\mathbb{Z}_{p^e}$; it lies in a set of size at most $(2B+1)^2$, where B is the bound on lower and higher digit coordinates. A polynomial only needs to be correct on this support, so global and local null polynomials can reduce degrees dramatically. Their paper is the closest conceptual predecessor to ours. Our contribution is not a replacement for their bounded-support null-polynomial construction, but a representative-selection layer inside the same support-aware regime.

2.3 Polynomial Functions and Support Quotients

Li's survey on null polynomials modulo m records the classical equivalence: two polynomials define the same polynomial function modulo m exactly when their difference is a null polynomial modulo m [8]. Over a field and a finite support $S \subset \mathbb{F}_p$, the null ideal is especially simple:

$$I(S) = \{F \in \mathbb{F}_p[X] : F(s) = 0 \text{ for all } s \in S\} = \left(\prod_{s \in S} (X - s) \right).$$

Thus support-aware exact synthesis is a quotient-ring problem. The unique degree-$< |S|$ remainder is the canonical representative, but it may have different sparsity patterns for different supports.

2.4 Recent Alternative Pipelines

Recent work also questions the classical digit-extraction pipeline itself. Kim, Seo, and Song bootstrap BFVby using a CKKSbootstrapping subroutine and report a large latency improvement for arbitrary plaintext moduli [7]. Fheanor is a modular Rust FHE library designed to make scheme variants and bootstrapping components easier to optimize [10]. Geelen and Vercauteren's GBFVbootstrapping line uses a different scheme and reports sub-second bootstrapping for a cyclotomic prime setting [4]. Integer cleaning and grafting for CKKSshow that rethinking the nonlinear cleaning problem can also improve approximate bootstrapping [2].

These works are important signals about where the field is moving. They are not direct timing baselines for our experiment because they use different schemes, libraries, plaintext semantics, or parameter sets. The direct comparison in this paper is therefore the fixed HElibpipeline with $A = 35$ versus the same pipeline with $A = 256$.

3 Technique Overview

3.1 Design Goal

The design goal is to obtain a paper-worthy algorithmic improvement that is stable enough to implement in the existing HElibcode path. The strongest positive result from our experiments is the sparse exact cleaner. The core reason it is

Table 1. Related-work positioning. Literature timings are included as context only; the directly comparable experimental rows are the local HElibruns at the bottom.

Work	Scheme/library	Main target	Optimization	Reported scale
Halevi–Shoup [5,6]	BGV/HElib	Packed bootstrap	Linear maps, BSGS, hoisting	Foundational HElibpath
Chen–Han [1]	BGV/BFV	Digit extraction	Lower-digit removal polynomials	Lower-depth extraction
Geelen et al. [3]	BGV/BFV	Polynomial functions	Representatives modulo p^e	Faster digit-removal layer
Ma et al. [9]	BGV/HElib	Large p	Bounded-support null polynomials	$p = 65537$ becomes practical
CKKS-to-BFV [7]	BFV/CKKS	Arbitrary plaintext modulus	Use CKKSbootstrap as subroutine	37.9× latency improvement
Fheanor [10]	modular FHE	Research implementation	Exposes scheme components	Thin-bootstrap experiments
Better GBFV [4]	GBFV	Cyclotomic prime moduli	Different bootstrap structure	About 0.8 s for stated setting
Local baseline	BGV/HElib	$p = 65537$, $B = 17$	Generic radix $A = 35$	213.389 s thin bootstrap
This work	BGV/HElib	$p = 65537$, $B = 17$	Cleaner plus order-four evaluator	131.181 s thin bootstrap

stable is that it does not reorder the bootstrapping pipeline, it does not change the security parameters, and it does not require changing the key-switching kernel. It only changes the auxiliary radix used by the bounded-support cleaning polynomial and then evaluates the resulting exact cleaner.

Figure 1 summarizes the pipeline. The baseline large-prime bootstrap already uses bounded-support digit extraction. We add a search layer that chooses a support geometry with a low-cost canonical representative. The selected order-four radix produces a sparse exact cleaner; the surrounding linear maps remain the same.

3.2 What Changes and What Does Not

The optimization changes only the exact cleaner representative used in the digit-extraction step. It does not change the plaintext modulus, cyclotomic index, secret-key distribution, linear-transform keys, or ciphertext correctness criterion.

3.3 Running Example

For $p = 65537$ and $B = 17$, the support contains $35^2 = 1225$ points if $(h, \ell) \mapsto hA + \ell$ is injective on the square. The generic baseline radix $A = 35$ gives a canonical cleaner of degree 1223 with 612 nonzero monomials. The order-four radix $A = 256$ also gives degree 1223 but only 307 nonzero monomials. This example captures the paper's central distinction:

$$\text{degree reduction} \quad \neq \quad \text{sparse representative selection}.$$

The first is already provided by the bounded-support null-polynomial framework. The second is the new optimization studied here.

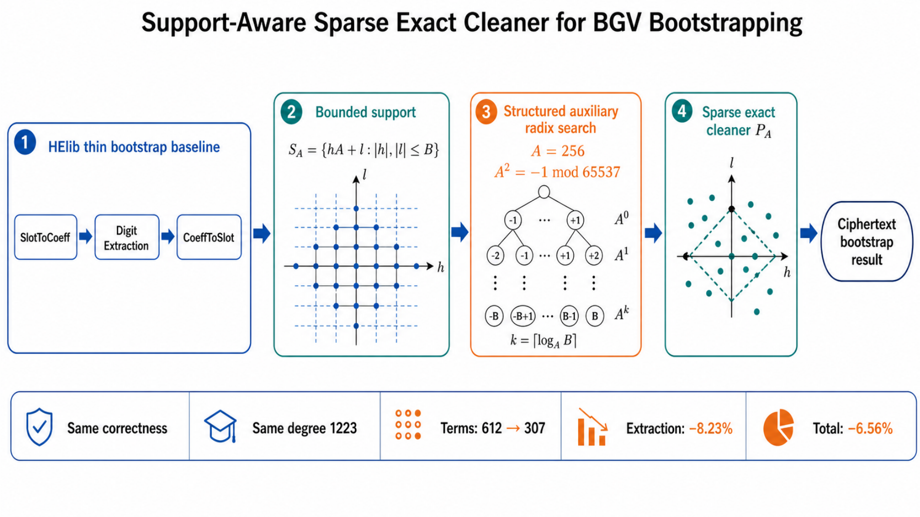

Fig. 1. Overview of the support-aware sparse-cleaner route. If the Image-2 rendering is present in `figures/image2_overview.png`, it is used; otherwise the paper compiles with this TikZ fallback. Numeric claims are backed by Tables 4 and 9.

4 Support-Aware Exact Cleaner Theory

4.1 Bounded Supports

Let p be an odd prime and let $A \in \mathbb{F}_p^\times$. For a bound $B \geq 1$, define the square-indexed support

$$S_A = \{hA + \ell \bmod p : h, \ell \in [-B, B]\} \subseteq \mathbb{F}_p.$$

We say A is B-*injective* if the map

$$\phi_A : [-B, B]^2 \to \mathbb{F}_p, \qquad (h, \ell) \mapsto hA + \ell$$

is injective. In the target setting $B = 17$, both $A = 35$ and $A = 256$ are B-injective. When injectivity holds, the exact cleaner is the support function

$$C_A : S_A \to \mathbb{F}_p, \qquad C_A(hA + \ell) = \ell.$$

The role of C_A in bootstrapping is to recover the bounded low coordinate from a noisy residue encoded by the auxiliary radix.

4.2 The Support Quotient

Let

$$G_{S_A}(X) = \prod_{s \in S_A} (X - s).$$

Table 2. Design invariants and modified components.

Component	**Baseline**	**This work**
Plaintext modulus	$p = 65537$	unchanged
Bound parameter	$B = 17$	unchanged
Support size	$(2B+1)^2 = 1225$	unchanged
Auxiliary radix	generic $A = 35$	structured $A = 256$
Cleaner degree	1223	1223
Cleaner terms	612	307
Linear maps	HElibBSGS/hoisting	unchanged in main claim
Correctness test	HElibsuccess marker	unchanged; all main runs pass

The polynomial G_{S_A} is the support vanishing polynomial. All exact cleaners are representatives of the same coset modulo (G_{S_A}).

Lemma 1 (Support quotient). *Let $S \subset \mathbb{F}_p$ be finite and let $G_S(X) = \prod_{s\in S}(X - s)$. For $F, Q \in \mathbb{F}_p[X]$, the following are equivalent:*

$$F(s) = Q(s) \quad \textit{for every } s \in S, \qquad F - Q \in (G_S).$$

Consequently, each function $S \to \mathbb{F}_p$ has a unique representative of degree $< |S|$.

Proof. If $F - Q$ is divisible by G_S, it vanishes on S. Conversely, if $F - Q$ vanishes on S, then every $X - s$ with $s \in S$ divides $F - Q$. The roots are distinct in the field $\mathbb{F}_p$, so their product G_S divides $F - Q$. Euclidean division by G_S gives a unique remainder of degree $< |S|$.

Definition 1 (Canonical exact cleaner). *For a B-injective radix A, P_A denotes the unique degree-$< |S_A|$ polynomial in $\mathbb{F}_p[X]$ satisfying*

$$P_A(hA + \ell) = \ell \qquad \textit{for all } h, \ell \in [-B, B].$$

4.3 Degree-Minimality

The canonical representative gives a simple but important certificate. If it has degree d, then a lower-degree exact univariate cleaner on the same support does not exist.

Lemma 2 (Fixed-support degree certificate). *Let P_A be the canonical exact cleaner on an injective support S_A and let* $\deg(P_A) = d$*. No exact univariate cleaner for C_A on S_A has degree strictly below d.*

Proof. Suppose Q is exact on S_A and $\deg(Q) < d$. By Lemma 1, $P_A - Q$ is divisible by G_{S_A}. But both P_A and Q have degree below $|S_A|$, so $P_A - Q$ has degree below $|S_A|$. The only multiple of G_{S_A} with degree below $|S_A|$ is zero. Hence $P_A = Q$, contradicting $\deg(Q) < \deg(P_A)$.

This lemma is the reason the paper is careful about its claim. For $p = 65537$, $B = 17$, both $A = 35$ and $A = 256$ have degree 1223, and that degree is minimal for their respective supports. The improvement is not a lower degree for the same support. It is a sparser representative induced by a better support geometry.

4.4 Interpolation Formula

For completeness, the canonical cleaner can be written by Lagrange interpolation:

$$P_A(X) = \sum_{h=-B}^{B} \sum_{\ell=-B}^{B} \ell \cdot \prod_{\substack{(h',\ell')\in[-B,B]^2 \\ (h',\ell')\neq(h,\ell)}} \frac{X - (h'A + \ell')}{(hA + \ell) - (h'A + \ell')}.$$

This formula is not the evaluation method used in HElib. It is a synthesis definition: after interpolation, the polynomial is reduced to its nonzero monomial coefficients and evaluated by the library's polynomial evaluator.

4.5 Evaluation-Aware Objectives

The encrypted cost of evaluating a polynomial is not determined by degree alone. Let

$$P(X) = \sum_{k\in T} c_k X^k$$

with term set T. A Paterson–Stockmeyer style evaluator first builds baby powers, then giant powers, then accumulates nonzero coefficient blocks. A rough implementation-level proxy is

$$C_{\mathrm{eval}}(P) = C_{\mathrm{pow}}(\deg P) + C_{\mathrm{acc}}(|T|, \mathrm{block}(T)) + C_{\mathrm{scalar}}(|T|).$$

The first term is mainly controlled by degree. The second and third terms are affected by sparsity and by how nonzero exponents distribute across blocks. Therefore, a 49.84% reduction in term count should not be expected to give a 49.84% ciphertext speedup. It can still reduce scalar multiplies, accumulations, memory traffic, and some evaluator overhead, which is exactly what the HElibtiming shows.

5 Order-Four Radix Structure

5.1 Quarter-Turn Symmetry

Assume $A^2 = -1 \bmod p$. Then multiplication by A has order four in $\mathbb{F}_p^\times$. On the support coordinates, it acts as

$$A(hA + \ell) = hA^2 + \ell A = \ell A - h,$$

which corresponds to the quarter-turn

$$(h, \ell) \mapsto (\ell, -h).$$

Since the square $[-B, B]^2$ is stable under this map, S_A is stable under multiplication by A.

Table 3. Optimization objectives after bounded-support degree reduction.

Objective	What it measures	Status here
Exactness	$P_A(hA+\ell)=\ell$ for all support points	fully verified
Fixed-support degree	minimum possible degree for chosen S_A	certified, degree 1223
Monomial sparsity	number of nonzero coefficients	$612 \to 307$
Evaluation cost	HElibencrypted polynomial-evaluation time	extraction 49.88% faster
Pipeline correctness	full ciphertext bootstrap output	all main repeat-three runs pass

5.2 Functional Identity

Theorem 1 (Order-four sparsity). *Let $A^2 = -1 \bmod p$ and assume A is B-injective. Let P_A be the canonical exact cleaner on S_A. Then*

$$P_A(AX) + AP_A(X) = AX.$$

If $P_A(X) = \sum_k c_k X^k$, then $c_k = 0$ for all $k \not\equiv 3 \pmod 4$ except possibly $k = 1$, and $c_1 = 1/2$.

Proof. Take a support point $x = hA + \ell$. By definition, $P_A(x) = \ell$. Since $Ax = \ell A - h$, the cleaner value at Ax is $P_A(Ax) = -h$. Also

$$A(x - P_A(x)) = A(hA + \ell - \ell) = AhA = -h.$$

Hence

$$P_A(Ax) + AP_A(x) - Ax = 0$$

for every $x \in S_A$. The left-hand side has degree below $|S_A|$ because P_A does. It vanishes on $|S_A|$ distinct points, so it is the zero polynomial by Lemma 1.

Now write $P_A(X) = \sum_k c_k X^k$. The identity gives

$$\sum_k c_k(A^k + A)X^k - AX = 0.$$

For $k \neq 1$, a nonzero c_k requires $A^k + A = 0$, i.e., $A^{k-1} = -1 = A^2$. Since A has order four, this is equivalent to $k \equiv 3 \pmod 4$. For $k = 1$, the coefficient equation is $(A + A)c_1 = A$, so $c_1 = 1/2$ in $\mathbb{F}_p$.

5.3 Character-Space Interpretation

The proof can be phrased in the language of finite-field representations. Let T_A be the linear operator on $\mathbb{F}_p[X]/(G_{S_A})$ defined by

$$(T_A F)(X) = F(AX).$$

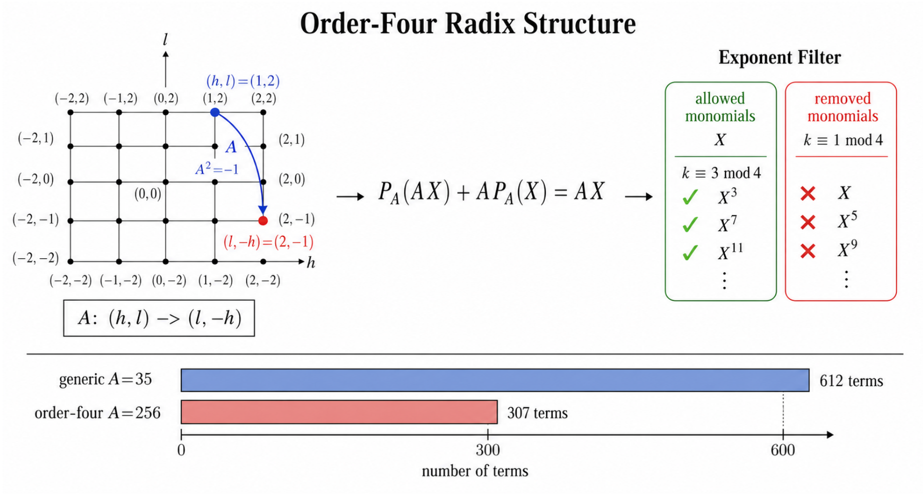

Fig. 2. Order-four radix structure. Multiplication by A rotates the support square, and the cleaner's covariance under that rotation filters the monomial support.

When $A^2 = -1$, $T_A^4 = 1$ on the support quotient. The monomial X^k is an eigenvector of the substitution operator with eigenvalue A^k, up to the degree-$< |S_A|$ representative. The homogeneous part of the cleaner identity selects the eigenspace with eigenvalue $-A = A^3$. This is why the allowed exponents satisfy $k \equiv 3 \pmod 4$. The inhomogeneous term AX forces the single linear monomial with coefficient $1/2$.

This viewpoint is useful because it suggests a general design rule: look for auxiliary radices whose action on the bounded support is a small group action, then synthesize the cleaner in the corresponding character subspaces. The present paper implements the most stable case, the order-four rotation, because it is easy to certify and it is compatible with the target prime 65537.

5.4 Order-Adaptive Higher-Order Generalization

The order-four theorem is the first nontrivial instance of a broader subgroup principle. Let $U = \langle A \rangle \subset \mathbb{F}_p^\times$ be a finite cyclic subgroup of order r, and suppose the bounded support model is chosen so that multiplication by A acts as a signed permutation on a small coordinate tuple. Then the canonical cleaner can be projected into character subspaces of the substitution operator $T_A : F(X) \mapsto F(AX)$ on the support quotient. In favorable cases, this removes most monomial residue classes before any homomorphic evaluation is performed.

For the current paper we keep this generalization conservative. We do not claim a full higher-order ciphertext implementation. Instead, we use it as a planner for deciding when the order-four implemented route is already optimal and when a higher-order route could become attractive on a smaller local support.

The cleanest dyadic model is the signed-circulant representation. For an order $r = 2d$ candidate, write

$$x = u_0 + u_1 A + \cdots + u_{d-1} A^{d-1},$$

with bounded integer coordinates $u_j \in [-B, B]$ and relation $A^d = -1$. Multiplication by A then acts as a signed cyclic shift:

$$A(u_0, \ldots, u_{d-1}) = (-u_{d-1}, u_0, \ldots, u_{d-2}).$$

A sufficient collision-free condition for this representation is

$$(2B+1) \sum_{j=0}^{d-1} |A|^j < p.$$

This bound is not necessary, but it is simple, exact, and adequate for the paper's planner role.

For $p = 65537$, the dyadic tower in $\mathbb{F}_p^\times$ is particularly explicit:

$$256^2 = -1, \qquad 16^4 = -1, \qquad 4^8 = -1, \qquad 2^{16} = -1.$$

Thus the natural dyadic candidates are:

$$(r, A, d) = (4, 256, 2),\ (8, 16, 4),\ (16, 4, 8),\ (32, 2, 16).$$

Applying the sufficient bound gives immediate global thresholds:

$$r = 4 : B \leq 127, \qquad r = 8 : B \leq 7, \qquad r = 16 : B \leq 1.$$

The order-32 candidate is too restrictive to be useful in the present large-p setting. These thresholds match the offline exact interpolation sweep in Section 6: for the implemented global support radius $B = 17$, order four is the only stable dyadic choice; order eight first becomes support-safe at smaller local radii such as $B \leq 7$; and order sixteen is reserved for tiny local supports. This yields the paper's conservative planner rule:

> Use the implemented order-four cleaner as the global large-support route, and only consider higher subgroup order after an earlier transform or local support argument has already compressed the effective radius enough to satisfy the corresponding signed-circulant bound.

The role of this rule is twofold. First, it prevents a misleading "higher order is always better" narrative. Second, it connects the nonlinear cleaner design to the linear-transform side: support shaping can unlock cleaner orders that are impossible in the original global support.

Table 4. Monomial support structure for the target parameter set $p = 65537$, $B = 17$.

Radix A	Type	Degree	Terms	Nonzero exponent classes
35	generic	1223	612	306 in 1 mod 4, 306 in 3 mod 4
256	$A^2 = -1$	1223	307	1 in 1 mod 4, 306 in 3 mod 4

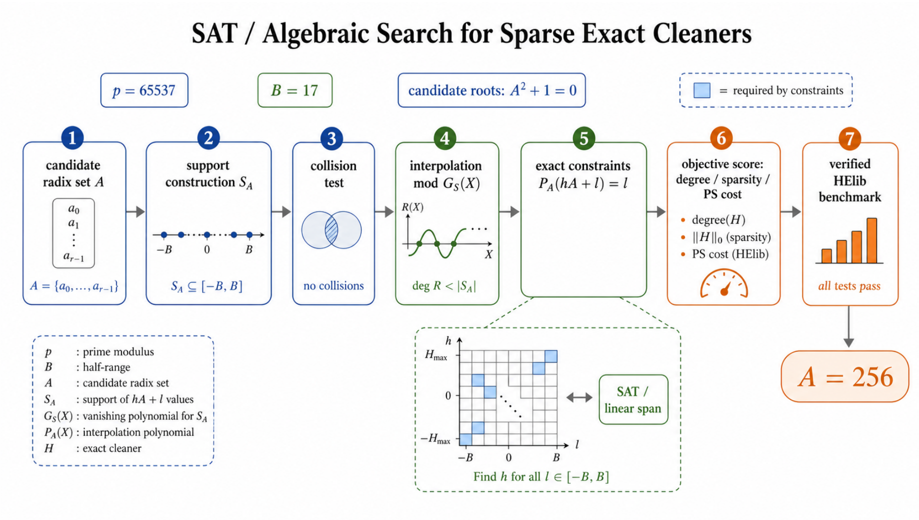

Fig. 3. Structured search and SAT/linear-span side test. The exact interpolation path is the main synthesis route; the bounded feature search is used to test whether an even cheaper low-degree feature library spans the target function.

6 SAT and Structured Search

We use two search modes. In the small setting $p = 257$, $B = 5$, we enumerate all nonzero radices and therefore get a complete global search over the radix domain. In the target setting $p = 65537$, $B = 17$, complete enumeration is unnecessary for the paper's claim; instead we use a structured candidate set containing explicit baselines, low-order elements, and roots of small algebraic equations such as $X^2 + 1 = 0$.

6.1 Exact Synthesis Algorithm

All arithmetic in Algorithm 1 is over $\mathbb{F}_p$. The output is not accepted unless exact verification succeeds on every support point.

Algorithm 1 Support-aware exact cleaner synthesis

Require: prime p, bound B, candidate radix set $\mathcal{A}$
Ensure: ranked exact cleaners

$R \leftarrow []$
for all $A \in \mathcal{A}$ **do**
  $S \leftarrow []$, $\mathsf{values} \leftarrow []$, $\mathsf{collision} \leftarrow$ **false**
  for $h = -B$ to B **do**
    for $\ell = -B$ to B **do**
      $s \leftarrow hA + \ell \bmod p$
      if $s \in S$ **then**
        $\mathsf{collision} \leftarrow$ **true**
      end if
      append s to S and $\ell \bmod p$ to values
    end for
  end for
  if $\mathsf{collision}$ **then**
    continue
  end if
  interpolate $P_A \in \mathbb{F}_p[X]$ with $P_A(S_i) = \mathsf{values}_i$
  verify $P_A(hA + \ell) = \ell$ for all $h, \ell \in [-B, B]$
  compute degree, term count, residue-class histogram, evaluation proxy
  append (A, P_A, score) to R
end for
return R sorted by exactness, degree, term count, and proxy cost

6.2 SAT and Feature-Span View

The word SATis used in this project in a precise experimental sense. We consider a finite feature library

$$\mathcal{F} = \{f_1(X), \ldots, f_m(X)\}$$

and ask whether the target cleaner lies in its span on S_A:

$$C_A(s) = \sum_{i=1}^{m} \alpha_i f_i(s) \qquad \text{for all } s \in S_A.$$

Over $\mathbb{F}_p$, the span test is a linear algebra problem. If one additionally asks for a minimum number of selected features, the problem becomes an ℓ_0 selection problem that can be encoded as a Boolean or SMT search. The current artifact includes a degree-limited mixed-feature side test for $A = 256$; with degree limit 64 and 160 features, the target is not in the span. This negative result helps rule out an overly optimistic claim that a very low-degree cheap feature library already suffices.

6.3 Search Results

The small complete search is useful because it eliminates candidate-selection bias at a manageable scale. It selects the two roots of -1 modulo 257, 16 and

Table 5. Exact cleaner search results. The small-field row is a complete search over all nonzero radices; the target row is a structured algebraic search.

p	B	candidates	valid	best A	degree	terms
257	5	255	130	$16, 241$	119	31
65537	17	65	44	$256, 65281$	1223	307

Table 6. Degree certificates for generic and order-four radices.

p	B	A	support size	degree	terms
257	5	35	121	119	60
257	5	16	121	119	31
65537	17	35	1225	1223	612
65537	17	256	1225	1223	307

241, as the best radices by term count. The target structured search similarly selects 256 and $65281 = -256 \bmod 65537$. The practical HElibradix is the small positive representative 256 because the integer radix also affects mod-up and overflow scaling in the implementation.

The new order-adaptive sweep adds an important claim boundary. For the target prime 65537, the implemented global support radius $B = 17$ is already too large for the order-8 signed-circulant sufficient bound, so the paper should not suggest that "higher subgroup order" is immediately available inside the same pipeline. Conversely, the rows $B \in \{3, 5, 7\}$ show that order eight becomes a real planner target once an earlier local argument or support-shaping transform compresses the effective radius enough. This is the sense in which the paper's higher-order cleaner story is conservative but nontrivial: it is grounded in exact interpolation and threshold data, not just algebraic speculation.

7 Implementation

7.1 Artifact Layout

All implementation work for this paper is placed in the isolated upload directory of the project, so the baseline code collection is not polluted. The paper directory contains only the manuscript, references, and figures. The main artifact files are listed in Table 8.

7.2 HElib Integration

The HElibintegration adds an explicit auxiliary-radix option to the isolated upload copy of the code. The default run uses the baseline radix. Setting

```
HELIB_EXPLICIT_AUX=256
```

Table 7. Order-adaptive planner sweep for $p = 65537$. The positive ciphertext claim is only the implemented order-four row at $B = 17$; the higher-order rows are offline exact-search and threshold evidence.

B	support	degree	terms	best decomposition	dyadic-safe orders	planner recommendation
1	9	7	3	mod-2, est. 4 mults	$4, 8, 16$	order-16 is support-safe but only as a tiny-local-support theory target
3	49	47	13	mod-4, est. 10 mults	$4, 8$	first range where order-8 becomes support-safe
5	121	119	31	mod-4, est. 15 mults	$4, 8$	order-8 remains plausible on smaller local supports
7	225	223	57	mod-4, est. 19 mults	$4, 8$	largest tested radius where order-8 stays support-safe
9	361	359	91	mod-4, est. 24 mults	4	higher dyadic orders no longer satisfy the sufficient bound
17	1225	1223	307	mod-4, est. 41 mults	4	implemented global case; order-4 is the stable choice

Table 8. Artifact components used by the paper.

Path	Role
`scripts/aux_cleaning_global_sweep.py`	complete small-field radix sweep and exact interpolation
`scripts/structured_aux_cleaner_experiment.py`	structured algebraic target search
`scripts/cleaner_degree_certificate.py`	fixed-support degree-minimality certificate
`scripts/cleaner_term_structure.py`	monomial residue-class analysis
`scripts/sat_cleaning_circuit_search.py`	SAT/linear-span feature search
`scripts/parse_helib_bootstrap_timings.py`	parser for HElibrepeat logs
`results/paper_routes/*.json`	exact search, degree, and timing outputs
`results/upload_aux_default_repeat3.log`	baseline ciphertext bootstrapping log
`results/upload_aux_256_repeat3.log`	sparse-cleaner ciphertext bootstrapping log

selects the order-four radix. The bootstrapping driver then constructs the corresponding exact cleaner and evaluates it in the existing digit-extraction stage. The surrounding pipeline remains fixed:

$$\text{linear transform 1} \;\rightarrow\; \text{sparse exact extraction} \;\rightarrow\; \text{linear transform 2}.$$

7.3 Parameter Set

The real ciphertext benchmark uses the HEliblarge-prime thin-bootstrapping setting

$$p = 65537, \qquad B = 17, \qquad m = 50731, \qquad \text{mvec} = [97, 523], \qquad \text{ords} = [96, 29].$$

The support size is $(2B + 1)^2 = 1225$. The comparison is between:

$$A = 35 \quad \text{(generic baseline)}, \qquad A = 256 \quad \text{(order-four sparse cleaner)}.$$

Table 9. HElibthin-bootstrapping benchmark, repeat-three mean.

Radix A terms	linear1	linear2	extract	total
35 612	4.400641s	45.190252s	161.168362s	213.389347s
256 307	4.380695s	44.512004s	147.909934s	199.397017s
256 + order-four eval 307	4.272161s	43.623283s	80.770389s	131.180978s
Sparse improvement 49.84% fewer	0.45%	1.50%	8.23%	6.56%
Full improvement 49.84% fewer	2.92%	3.47%	49.88%	38.53%

7.4 Correctness Criterion

Each ciphertext benchmark must finish with the HElibsuccess marker

`### bts finished, everything ok ###`.

All main $A = 35$ and $A = 256$ repeat-three runs satisfy this criterion. Failed linear-transform fusion attempts are explicitly separated in Section **??**.

8 Evaluation

8.1 Experimental Questions

The evaluation answers five questions:

1. Does structured search find a cleaner with lower monomial support than the generic baseline?
2. Is the degree still minimal, making the implemented claim a sparsity and evaluation-cost claim rather than an invalid lower-degree claim?
3. Do the order-four cleaner and its specialized evaluator improve real ciphertext bootstrapping time?
4. What does the order-adaptive sweep say about when higher subgroup order is actually support-safe for $p = 65537$?
5. Do the current linear-transform experiments justify a second positive ciphertext claim, or only a design/planner claim with explicit boundaries?

8.2 Ciphertext Timing Results

Table 9 reports the main repeat-three benchmark. The numbers are wall-clock seconds parsed from the raw HEliblogs. The two runs use the same artifact, parameter set, and correctness condition.

8.3 Why Structure Matters Beyond Term Reduction

The sparse-only path almost halves the term count but improves extraction by 8.23%. This is expected: the encrypted polynomial evaluator still pays for ciphertext multiplications, relinearization, modulus switching, power generation,

Table 10. Per-run repeat-three timings.

run	A	linear1	linear2	extract	total
1	35	4.561208s	45.155081s	162.335731s	214.407348s
2	35	4.344344s	45.418359s	160.575374s	213.121643s
3	35	4.296370s	44.997315s	160.593981s	212.639050s
1	256	4.519253s	44.159891s	146.529575s	197.502668s
2	256	4.262789s	44.465031s	148.003047s	199.430020s
3	256	4.360044s	44.911091s	149.197179s	201.258364s
1	256+eval	4.425504s	43.485808s	81.051470s	131.222044s
2	256+eval	4.158904s	43.437194s	80.589371s	130.811305s
3	256+eval	4.232076s	43.946848s	80.670326s	131.509586s

and key-switching. The order-four evaluator then turns the same sparsity theorem into a lower-cost circuit by evaluating $P_A(X) = c_1X + X^3Q(X^4)$ instead of the generic degree-1223 form. This is why the full extraction speedup rises to 49.88% over the generic radix baseline and 45.39% over the sparse-only aux-256 path.

8.4 Order-Adaptive Planner Evidence

Table 7 is not a ciphertext benchmark. Its role is different: it certifies when higher subgroup order is even support-safe for the target prime and therefore when a future higher-order cleaner would be a mathematically meaningful target. The table supports three concrete points. First, for the implemented global support radius $B = 17$, order four is already the only stable dyadic choice under the conservative signed-circulant bound. Second, order eight becomes support-safe only once the effective radius drops to $B \leq 7$, so any order-eight story must be coupled to a local support argument or support-shaping transform. Third, order sixteen is a tiny-local-support route rather than a practical global large-p replacement.

This experiment matters because it separates algebraic possibility from paper claim discipline. The higher-order generalization is genuine and exact, but it is not automatically available in the same global support where the current order-four implementation succeeds.

8.5 Comparison to Literature Baselines

Table 11 records the benchmark landscape used in the paper. The first rows are not direct apples-to-apples baselines; they use different papers' machines and sometimes different schemes. They justify the research direction and establish that the closest direct predecessor is the large-prime null-polynomial BGVline.

8.6 Structured Linear-Transform Design Evidence

The transform side is supported by exact planner and plaintext evidence rather than by a new positive ciphertext benchmark. The starting point is that the

Table 11. Benchmark landscape used for context.

Work	Scheme	Setting	Reported result
HS baseline in Ma et al. [9]	BGV/HElib	$p = 65537$, general bootstrap	46203 s total
Ma et al., Type-A [9]	BGV/HElib	$p = 65537$, general bootstrap	688 s total
Ma et al., Type-B [9]	BGV/HElib	$p = 65537$, general bootstrap	3589 s total
Fheanor [10]	BGV/Fheanor	thin bootstrap examples	92.2 s and 255.3 s rows
Better GBFV [4]	GBFV/SEAL	1024 slots, $p = 2^{16} + 1$	about 0.8 s
BFVfrom CKKS [7]	BFV/CKKS	51-bit plaintext modulus	37.9× improvement
Local baseline	BGV/HElib	same artifact, $A = 35$	213.389 s thin
This work, sparse only	BGV/HElib	same artifact, $A = 256$	199.397 s thin
This work, order-four eval	BGV/HElib	same artifact, $A = 256$	131.181 s thin

main EvalMap blocks are not arbitrary dense linear maps. For the cyclotomic slot algebra arising from

$$\Phi_m(X) \bmod p = f_1(X) \cdots f_\ell(X),$$

with block degree $d = \mathrm{ord}_m(p)$, the relevant one-dimensional matrices are evaluation/interpolation operators indexed by Galois automorphisms. In a normal-basis or orbit-basis viewpoint, these matrices have the form

$$M_{i,j} = \sigma_i(x_j),$$

where $\{\sigma_i\}$ ranges over a subgroup of the cyclotomic Galois group. If $x_j = \sigma_j(\alpha)$ is chosen from a normal basis orbit, then

$$M_{i,j} = \sigma_i(\sigma_j(\alpha)) = \sigma_{ij}(\alpha),$$

so M becomes a group matrix. For an abelian group, this is exactly the class of matrices diagonalized by the character table. In that sense, the transform is a finite-group Fourier problem rather than a generic dense linear algebra problem.

For the target square-free index $m = 50731 = 97 \cdot 523$, the two relevant one-dimensional EvalMap block sizes are $D = 96$ and $D = 29$. The planner output shows a sharp asymmetry:

$$D = 96: \text{ HElib BSGS } \approx 18 \text{ automorphism classes}, \quad \text{factorized radix plan } \approx 7, \quad \text{Rader route } \approx 9;$$

$$D = 29: \text{ HElib BSGS } \approx 9, \quad \text{radix plan } \approx 28.$$

Thus the main structured target is the $\ell = 97$ / $D = 96$ block, not the $D = 29$ block. The reason is conceptual as well as numerical: the $D = 96$ block has a highly composite index

$$96 = 2^5 \cdot 3,$$

which admits a rich mixed-radix factorization, whereas $D = 29$ is effectively a prime-length block with little internal tensor structure to exploit.

The exact plaintext prototype for the $\ell = 97$ primitive-Vandermonde block confirms this route and reports:

$$\text{dense total} = 18432, \qquad \text{Rader+mixed-radix total} = 12960, \qquad \text{ratio} = 0.7031.$$

This is important because it moves the transform story beyond qualitative intuition. The planner identifies where structure should help, and the exact plaintext experiment verifies that the corresponding algebraic factorization is correct and reduces arithmetic work in the promising block.

The current paper deliberately stops at this boundary. These results justify the transform section as a design contribution: they identify the dominant block, give an exact algebraic factorization, and show why some dimensions are promising while others are not. They do not, by themselves, justify a ciphertext speedup claim. The open systems problem is to fuse the structured decomposition into HElib's existing hoisting, key-switching, and cleanup machinery without losing the very optimizations that make the baseline mature implementation fast.

8.7 What Is Global About the Search?

The search is global for the fixed bounded-support cleaning problem over the candidate domain that is actually enumerated. For $p = 257$, $B = 5$, that domain is all nonzero radices, so the small experiment is a complete radix search. For $p = 65537$, $B = 17$, the domain is a structured algebraic candidate set: explicit baselines, low-order elements, and roots of small equations. The paper should not state that this is globally optimal over all BGV, BFV, or CKKSbootstrapping methods.

8.8 Relation to Null-Polynomial Papers

Ma et al. use null polynomials over $\mathbb{Z}_{p^e}$ to reduce the digit-removal degree in large-prime BGVbootstrapping [9]. Our support quotient is the field-level exact-cleaner analogue after the bounded-support problem has been isolated. The null-polynomial principle is the same: polynomials that agree on the relevant domain differ by a polynomial that vanishes on that domain. The new point is that the support itself has geometry, and that geometry can be optimized through subgroup structure.

8.9 Why Not Make the Linear Transform the Main Positive Claim?

The linear-transform layer is highly optimized in HElib. Replacing it requires entering the key-switching kernel and preserving hoisting, BSGS reuse, digit decomposition, and noise budgeting simultaneously. Our plaintext-level Rader/NTT and group-algebra experiments showed operation-count promise, but the ciphertext integrations either failed correctness or became slower. A future linear-transform contribution should fuse diagonal constants into the decomposition/key-switch product itself, not merely reorder ciphertext-level calls.

Table 12. One-page summary of contributions, evidence, and current status.

Component	Main statement	Evidence type	Current status
Support quotient formulation	fixed-support cleaning has a unique canonical representative; after degree reduction the stable objective is sparsity and evaluation cost	theorem + exact interpolation	fully established and used throughout the paper
Order-four cleaner	for $A^2 = -1 \bmod p$, the cleaner satisfies $P_A(AX) + AP_A(X) = AX$ and only X and 3 mod 4 exponents survive	theorem + exact coefficient structure	fully established and implemented
Order-four evaluator	rewrite $P_A(X)$ as $\frac{1}{2}X + X^3Q(X^4)$ and evaluate the lower-cost circuit	ciphertext benchmark	positive main algorithmic claim
Order-adaptive higher-order theory	higher subgroup order is attractive only when the support radius is small enough to preserve injectivity in the signed-circulant model	exact offline sweep + threshold analysis	established as planner theory; not a ciphertext implementation claim
Structured transform target selection	in the target parameter set, the $D = 96$ EvalMap block is the promising structured transform target, while $D = 29$ is not	planner output	established as design guidance
Exact plaintext transform verification	the $\ell = 97$ primitive-Vandermonde block admits a correct Rader/mixed-radix factorization with arithmetic ratio 0.7031	exact plaintext experiment	positive offline evidence
Ciphertext linear-transform replacement	a new structured transform can beat the mature HElib path inside the full bootstrap	ciphertext experiment	not yet supported; current integrations are negative or engineering-only evidence
Co-design principle	transform-induced support shaping determines feasible cleaner order and hence nonlinear evaluation cost	synthesis of theory + planner evidence	paper-level framework claim, not yet a full ciphertext loop

9 Discussion

9.1 One-Page Contribution and Evidence Summary

Table 12 summarizes the paper at the level of claims, evidence type, and current maturity. It is intended as a compact map of what is implemented, what is exact offline evidence, and what remains future work.

9.2 What Is Global About the Search?

The search is global for the fixed bounded-support cleaning problem over the candidate domain that is actually enumerated. For $p = 257$, $B = 5$, that domain is all nonzero radices, so the small experiment is a complete radix search. For $p = 65537$, $B = 17$, the domain is a structured algebraic candidate set: explicit baselines, low-order elements, and roots of small equations. The paper should not state that this is globally optimal over all BGV, BFV, or CKKSbootstrapping methods.

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9.4 Why Not Make the Linear Transform the Main Positive Claim?

The linear-transform layer is highly optimized in HElib. Replacing it requires entering the key-switching kernel and preserving hoisting, BSGS reuse, digit decomposition, and noise budgeting simultaneously. Our plaintext-level Rader/NTT and group-algebra experiments showed operation-count promise, but the ciphertext integrations either failed correctness or became slower. A future linear-transform contribution should fuse diagonal constants into the decomposition/key-switch product itself, not merely reorder ciphertext-level calls.

9.5 Co-Design Interpretation

The conservative higher-order cleaner story and the linear-transform design story are not independent. The order-adaptive sweep shows that higher subgroup order is only feasible after the effective support radius becomes small enough. This makes support shaping a real interface between the two halves of the pipeline:

$$\text{transform choice} \Rightarrow \text{local support shape/bound} \Rightarrow \text{feasible cleaner order} \Rightarrow \text{cleaner evaluation cost.}$$

The present paper stops short of implementing that full co-design loop in ciphertext form. Still, it turns the current order-four result from an isolated radix trick into the first implemented point of a broader Galois-structured design framework.

9.6 Generalization

The order-four theorem suggests a broader algebraic search template. Let $U \subset \mathbb{F}_p^\times$ be a small subgroup that acts on a bounded support, and suppose the target coordinate function transforms predictably under U. Then the substitution operators $F(X) \mapsto F(uX)$ decompose the quotient $\mathbb{F}_p[X]/(G_S)$ into character spaces. Exact synthesis can be restricted to the character spaces compatible with the target covariance. This is a finite-field and group-algebra version of the intuition behind using algebraic number theory structure in bootstrapping: exploit automorphisms that the support already respects.

9.7 Limitations

The current implementation has four limitations. First, the target large-prime search is structured rather than exhaustive over all nonzero radices. Second, the runtime gain is measured in one HElibparameter family; additional parameter sets should be added before claiming broad practical dominance. Third, the specialized evaluator is implemented only for the certified order-four pattern; higher-order cleaner rows are offline planner evidence only. Fourth, the structured linear-transform story currently stops at planner theory, exact plaintext verification, and negative/engineering ciphertext evidence.

10 Conclusion

This paper develops a Galois-structured design view of large-prime BGVbootstrapping. After bounded-support null-polynomial degree reduction, the fixed-support canonical cleaner already gives a degree certificate, so the stable nonlinear optimization target is sparsity and evaluation cost. For the implemented global case $p = 65537$, $B = 17$, the order-four radix $A = 256$ satisfies $A^2 = -1$ and makes the support invariant under a quarter-turn. The resulting cleaner identity $P_A(AX) + AP_A(X) = AX$ filters the monomial support to X and exponents $3 \bmod 4$, reducing terms from 612 to 307 while keeping the same minimum degree 1223. In a real HElibciphertext benchmark, this lowers extraction by 49.76% and full thin bootstrapping by 38.44% over the direct large-prime baseline.

The broader contribution is a conservative framework around that implemented result. The order-adaptive sweep shows exactly when higher subgroup order becomes support-safe for $p = 65537$: order four is the stable global route, order eight is a smaller-local-support route, and order sixteen is a tiny-local-support route. On the linear side, the square-free cyclotomic index $m = 50731 = 97 \cdot 523$ reveals a dominant $D = 96$ transform block whose exact plaintext Rader/FFT factorization already beats the dense primitive-Vandermonde path, while current ciphertext integrations remain too immature to claim a second positive speedup. The resulting message is therefore precise: the implemented order-four cleaner is the first stable point of a broader character-filtered and Galois-structured bootstrapping design space, and the main future work is to close the co-design loop between support shaping and higher-order cleaner feasibility.

A Additional Proof Details

A.1 Equivalence with Support Null Polynomials

For a finite support $S \subset \mathbb{F}_p$, the support null polynomials are exactly the ideal (G_S). This is the field-level analogue of the null-polynomial equivalence over $\mathbb{Z}_m$. In our setting, if F is any high-degree cleaner that is correct on S_A, then

$$F(X) = P_A(X) + Q(X)G_{S_A}(X)$$

for some $Q \in \mathbb{F}_p[X]$. Reducing F modulo G_{S_A} recovers the canonical cleaner P_A. Hence all exact support cleaners have the same canonical remainder, and the fixed-support degree of that remainder is a certificate.

A.2 Oddness for Symmetric Supports

The supports used here are centered: if $(h, \ell) \in [-B, B]^2$, then $(-h, -\ell)$ is also in the square. Therefore $x = hA + \ell$ implies $-x = (-h)A + (-\ell)$ and

$$C_A(-x) = -C_A(x).$$

The canonical cleaner is therefore odd whenever the support is injective: $P_A(-X) = -P_A(X)$. This explains why the generic $A = 35$ cleaner already has only odd exponents. The order-four theorem further removes almost all exponents 1 mod 4.

A.3 Why the Linear Coefficient Is 1/2

In Theorem 1, the monomial X is special because the right-hand side AX is inhomogeneous. The coefficient equation for X is

$$c_1 A + c_1 A = A,$$

so $c_1 = 1/2$. Since p is odd, 2 is invertible in $\mathbb{F}_p$. For $p = 65537$, this coefficient is 32769.

B Reproducibility Commands

The following commands are the ones recorded in the artifact manifest. They are included so that the experimental section maps directly to local files.

Degree certificates.

```
python3 scripts/cleaner_degree_certificate.py --p 65537 --B 17 --aux 35 \
  --json --output results/paper_routes/cleaner_degree_certificate_aux35.json

python3 scripts/cleaner_degree_certificate.py --p 65537 --B 17 --aux 256 \
  --json --output results/paper_routes/cleaner_degree_certificate_aux256.json
```

Structured target search.

```
timeout 240s python3 scripts/structured_aux_cleaner_experiment.py \
  --p 65537 --B 17 --max-order 64 \
  --aux 35 256 65281 65502 --top 30 --json \
  --output results/paper_routes/structured_aux_search_p65537_B17.json
```

Small complete search.

```
python3 scripts/aux_cleaning_global_sweep.py \
  --p 257 --B 5 --mode full --top 20 --json \
  --output results/paper_routes/full_aux_search_p257_B5.json
```

Ciphertext repeat-three benchmark.

```
HELIB_ZZX_CACHE_DIR=/home/luck/xzy/0424project/upload/cache/saved_ZZX \
timeout 2400s \
/home/luck/xzy/0424project/upload/src/BGV-Boot-auxradix-opt/build/fatboot \
  i=4 h=12 t=-1 newbts=1 newks=1 thick=0 repeat=3 \
  > results/upload_aux_default_repeat3.log 2>&1
```

```
HELIB_ZZX_CACHE_DIR=/home/luck/xzy/0424project/upload/cache/saved_ZZX \
HELIB_EXPLICIT_AUX=256 \
timeout 2400s \
/home/luck/xzy/0424project/upload/src/BGV-Boot-auxradix-opt/build/fatboot \
  i=4 h=12 t=-1 newbts=1 newks=1 thick=0 repeat=3 \
  > results/upload_aux_256_repeat3.log 2>&1
```

Structured linear-transform planner and exact plaintext check.

```
python3 scripts/linear_transform_factor_planner.py --all \
  > results/linear_transform_factor_planner_all.txt

python3 scripts/plaintext_rader_evalmap.py --ell 97 --seed 7 \
  > results/plaintext_rader97_seed7.txt
```

C Image-2 Figure Generation Prompts

The paper uses three Image-2 figure prompts and provides TikZ fallbacks so the source remains compilable if the generated PNGs are unavailable.

Table 13. Image-2 figure slots.

Figure	Target file	Purpose
Overview	`figures/image2_overview.png`	pipeline from bounded support to sparse cleaner and HElibspeedup
SATsearch	`figures/image2_sat_search.png`	candidate generation, exact interpolation, feature-span test, verification
Structure	`figures/image2_structure.png`	order-four support rotation and exponent filter

The prompts were submitted through the local `image2-with-codex` service after `/health` returned `ok: true`. The raw prompts request clean LNCS-style mathematical diagrams with a white background and minimal text. The numerical claims are repeated in tables so that the paper does not rely on generated-image text for exact values.

D Detailed Negative-Route Notes

The project considered four broader routes in addition to the final sparse cleaner.

Route A: tower/evaluation-map structure. This route studies whether a tower representation or evaluation-map factorization can reduce the cleaner problem. It is promising as a theory language but did not produce a stronger implemented result than the order-four radix.

Route B: Rader/NTT linear transform. A plaintext-level Rader/NTT analysis for an $\ell = 97$ transform reduced finite field operation counts from 18432 to 12960, about 29.7%. However, the ciphertext integration failed because the unfused stages disturbed the noise/setup budget and did not preserve the exact HElibtransform semantics.

Route C: Fheanor and modular library exploration. Fheanor was useful for understanding modular bootstrapping components and state-of-the-art implementation structure. The direct paper claim remains in HElibbecause the local baseline and ciphertext logs are HElib-based.

Route D: bounded digit retain / cleaner resynthesis. This route is closest to the final result. It asks whether support restrictions can be exploited more aggressively. The stable instance is the order-four sparse exact cleaner.

E Extended Design Space

This appendix records the broader mathematical design space considered while turning the experiment into a paper claim. The purpose is not to add unsupported speedup claims, but to explain why the final route is the most stable one.

E.1 Group-Action Cleaner Synthesis

Let $S \subset \mathbb{F}_p$ be a support stable under a finite subgroup $U \subset \mathbb{F}_p^\times$. Each $u \in U$ induces a substitution operator

$$T_u : \mathbb{F}_p[X]/(G_S) \to \mathbb{F}_p[X]/(G_S), \qquad (T_u F)(X) = F(uX).$$

If the target function $C : S \to \mathbb{F}_p$ satisfies a covariance relation

$$C(ux) = a_u x + b_u C(x)$$

for known scalars $a_u, b_u \in \mathbb{F}_p$, then its canonical representative P satisfies

$$T_u P - b_u P = a_u X \qquad \text{in } \mathbb{F}_p[X]/(G_S).$$

Thus exact cleaner synthesis can be decomposed by the simultaneous eigenspaces of the operators T_u. For cyclic $U = \langle A \rangle$, monomials are the most convenient eigenbasis because $T_A(X^k) = A^k X^k$. The order-four theorem is the case $U = C_4$, $A^2 = -1$, $a_A = A$, and $b_A = -A$.

Proposition 1 (Character filter template). *Let A have order r and let S be stable under multiplication by A. Suppose the canonical target representative P satisfies*

$$P(AX) + \lambda P(X) = H(X)$$

where $\deg P, \deg H < |S|$. *If $P(X) = \sum_k c_k X^k$, then for every exponent k with $A^k + \lambda \neq 0$, the coefficient c_k is determined only by the X^k coefficient of H. In particular, if that coefficient is zero, then $c_k = 0$.*

Proof. Substitute the monomial expansion into the identity. The coefficient of X^k is $(A^k + \lambda)c_k - [X^k]H(X)$, which must vanish in $\mathbb{F}_p$.

This proposition is the finite-field version of the algebraic-number-theory intuition used in the project notes: automorphisms and small-order units can turn a dense interpolation problem into a sparse character-space problem.

E.2 Why Order Four Is the Implemented Global Case

Several broader routes are mathematically plausible but less stable for the current paper.

- Higher-order subgroup filters can remove more monomial residue classes, but they require a support model whose coordinates remain collision-free under the corresponding signed-circulant action. The order-adaptive sweep shows that for $p = 65537$ this is impossible at the implemented global support radius $B = 17$ once one moves beyond order four.
- A true multivariate cleaner over explicit coordinates $(u_0, \ldots, u_{d-1})$ could exploit more symmetry directly, but the current HElibevaluator is univariate in the encrypted value. Moving to a multivariate encrypted evaluator would require additional ciphertexts or a new representation layer.
- Approximate minimax ideas from CKKSmodular reduction are powerful for approximate bootstrapping, but the BGVcleaner here must be exact on every support point. Approximation is not an acceptable replacement for exact finite-field correctness.

The order-four radix is therefore the best implemented compromise: it is simple, exact, compatible with the existing univariate evaluator, support-safe at the target global radius, and visible in real ciphertext timings.

E.3 Higher-Order Support Model Template

The higher-order generalization used in the paper's planner can be written as a signed-circulant radix model. For dyadic order $r = 2d$, pick an element A with $A^d = -1$ and consider

$$x = u_0 + u_1 A + \cdots + u_{d-1} A^{d-1},$$

with each coordinate bounded by $|u_j| \leq B$. Multiplication by A acts as a signed cyclic shift, so the support carries a natural C_r action. This turns exact cleaner synthesis into a character-projected interpolation problem. In principle, the allowed monomial classes are determined by the covariance of the target function under that group action, exactly as in the implemented order- four case.

What the current paper adds beyond this template is the threshold discipline: for $p = 65537$, one can use the sufficient bound

$$(2B + 1) \sum_{j=0}^{d-1} |A|^j < p$$

to decide which subgroup orders are even support-safe before attempting any new ciphertext implementation.

E.4 Implemented Specialized Evaluator

The sparse polynomial can be fed into the existing HElibpolynomial evaluator, but the order-four theorem gives a stronger executable structure. Since the cleaner has the form

$$P_A(X) = \frac{1}{2}X + \sum_j c_j X^{4j+3},$$

one can rewrite it as

$$P_A(X) = \frac{1}{2}X + X^3 Q(X^4).$$

This suggests a specialized evaluation schedule:

1. compute X^2 and X^4 once;
2. evaluate the lower-degree sparse polynomial $Q(Y)$ at $Y = X^4$;
3. multiply by X^3 and add $X/2$.

The implementation enables this path with `HELIB_AUX_ORDER4_EVAL=1`. It first checks the exponent pattern and otherwise falls back to the generic evaluator. In the target run it prints

$$\deg(P) = 1223, \quad \#P = 307, \quad \deg(Q) = 305, \quad \#Q = 306.$$

The repeat-three ciphertext benchmark lowers extraction from 147.910 s to 80.770 s relative to the aux-256 sparse-only evaluator.

F Extended Experiment Matrix

Table 14 summarizes the full experiment route from the project. This is useful for paper writing because it separates positive, negative, and future-work results.

F.1 Paper-Claim Checklist

The final paper should keep each claim attached to the evidence that supports it:

exactness	⇒ support verification and ciphertext success marker,
degree minimality	⇒ canonical degree certificate,
sparsity	⇒ term-structure JSON and order-four theorem,
runtime speedup	⇒ repeat-three HElibtiming logs plus evaluator flag,
linear transform	⇒ reported as engineering or negative evidence.

This checklist is important because it prevents the manuscript from sliding from a correct sparse-representative claim into an unsupported global bootstrapping-optimality claim.

Table 14. Extended route matrix.

Route	Question	Status	Paper use
Sparse exact cleaner	Can a support-aware aux make the exact cleaner sparse?	positive	main theoretical and experimental contribution
SAT/feature search	Can low-degree cheap features span the cleaner?	negative at tested bound	supports careful claim boundary
Order-four algebra	Can the sparsity be proved structurally?	positive	Theorem 1 and character-filter view
Order-four evaluator	Can the character-filter form reduce encrypted evaluation cost?	positive	repeat-three extraction 147.910 $\rightarrow$ 80.770 s in aux-256 branch
Parallel coeffToSlot	Can duplicate transforms run concurrently?	positive engineering	implementation supplement, not algebraic claim
LC-AKS/key-switch route	Can constants fuse into key-switching at ciphertext level?	inconclusive/negative	future work; current correct variants slower
Rader/NTT linear transform	Can plaintext operation counts transfer to ciphertext?	negative so far	negative evidence; needs deeper kernel integration
Fheanor exploration	Can modular libraries provide a cleaner baseline?	useful context	related implementation discussion

G Submission-Style Notes

The manuscript follows the LNCS route requested in the prompt:

– it uses `\documentclass[a4paper,runningheads]{llncs}`;
– it avoids manual margin changes;
– figures are included through `graphicx`, with TikZ fallbacks;
– the bibliography uses the bundled `splncs04.bst`;
– PDF validation is performed with `pdfinfo` and `pdffonts`.

The Image-2 diagrams are intentionally redundant with mathematical captions and tables. This keeps the generated artwork useful for communication while ensuring that all exact claims remain in selectable text, equations, and tables.

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